

# PAPER\_476 — DPM Pre-Big Bang 26-Sphere Birth Model

Star-Magic Unified Quantum Field Framework (UQFF) Whitepaper Series Copyright  
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## Abstract

The Dimensional Progenitor Model (DPM) describes the pre-Big Bang quantum state as 26 spherical shells embedded in a unit hypersphere, each encoding a binary quantum level. The [SCm] field provides  $10^{-2}$  J of binding energy across the shell structure, while the [UA] field decays exponentially during inflation. This paper derives the 26-sphere geometry, the energy budget, the resonance factor linking DPM states to gravitational output, and the connection to the 26-dimensional framework pervading UQFF physics.

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## 1. Introduction

The DPM answers the question: what was the pre-Big Bang quantum state? The UQFF answer is that the primordial spacetime comprised 26 spherical shells on a unit hypersphere, with each shell representing one quantum electromagnetic level. This is not merely numerology — the 26 emerges from the dimensionality of bosonic string theory and provides the natural binary encoding for the UQFF field hierarchy.

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## 2. The 26-Sphere Geometry

### 2.1 Unit Sphere Equation

Each sphere  $k \in \{1, \dots, 26\}$  satisfies:

$$(x - h_k)^2 + (y - k_k)^2 + (z - l_k)^2 = r_k^2$$

where  $(h_k, k_k, l_k)$  are the center coordinates of sphere  $k$  on the unit hypersphere  $S^2$  (25-sphere boundary of 26D space), and  $r_k$  is the sphere's radius.

In the canonical DPM configuration: - **All spheres initially coincide** at the origin ( $h=k=l=0$ ,  $r=1$  normalized) - **Inflation separates them** along independent dimensional axes - **26 directions** = 26 bosonic degrees of freedom

### 2.2 26-Sphere Volume Sum

$$V_{DPM} = \sum_{k=1}^{26} \frac{4}{3} \pi r_k^3$$

At Big Bang:  $r_k = r_{\text{Planck}} = 1.616 \times 10^{-35} \text{ m} \rightarrow V_{DPM} = 26 \times (4/3) r_P^3 = 7.24 \times 10^{-105} \text{ m}^3$

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### 3. Energy Budget

#### 3.1 [SCm] Binding Energy

The superconducting medium provides binding energy across the 26-shell structure:

$$E_{SCm} = [SCm]_{amount} \times A_{CP,massive}$$

where  $A_{CP,massive}$  is the massive compact-Planck coupling area.

**Canonical value:**  $E_{SCm} \approx 10^{42}$  J

This is approximately the gravitational binding energy of the Milky Way, placing the DPM energy scale at galactic energies — consistent with UQFF's galactic calibration.

#### 3.2 [UA] Energy Decay

During inflation, the [UA] field decays:

$$E_{UA}(t) = E_{UA,0} \times e^{-\gamma t}$$

with  $0.001 \text{ s}^{-1}$  (inflationary damping rate) and  $E_{UA,0} = 10^4$  J (initial energy).

At  $t_{\text{inflation}} = 10^{-32}$  s:  $E_{UA} \approx 10^{45} \times e^{-10^{-35}} \approx 10^{45}$  J (barely decayed)

At  $t = 10$  Gyr:  $E_{UA} \approx 10^{45} \times e^{-\gamma \times 3 \times 10^{17}} \approx \text{negligible} \rightarrow \text{present vacuum density } \rho_{\text{vac\_UA}} = 7.09 \times 10^{-3} \text{ J/m}^3$

#### 3.3 Resonance Factor

DPM states couple to gravity through:

$$R_{DPM} = \frac{GM}{r^2} \times q_{Higgs} \times H_{support} \approx 10^{-11} \text{ (normalized)}$$

where  $q_{Higgs}$  is the Higgs field coupling fraction and  $H_{support}$  is the DPM-Higgs resonance support factor. At  $r = 1$  m,  $M = 1$  kg:  $R = G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ .

## 4. Binary Encoding in 26 Levels

### 4.1 Level Classification

The 26 spheres divide into two groups:

| Group     | Levels | State                   | Energy                              |
|-----------|--------|-------------------------|-------------------------------------|
| +½ states | 1–13   | Low-energy EM channel   | $E_{SCm}/2$                         |
| –½ states | 14–26  | High-energy SC barriers | $E_{SCm} \times H_{\text{barrier}}$ |

**Physical interpretation:** -  $+1/2$  **states:** Standard electromagnetic photons + UQFF Ug1–Ug4 fields -  $-1/2$  **states:** High-energy superconducting barriers that prevent quantum decoherence

$$\Delta E_{barrier} = E_{SCm} \times H_{SCm} \approx 10^{42} \times 0.99 \approx 9.9 \times 10^{41} \text{ J}$$

## 4.2 26D Binary Coupling

Each of the 26 levels provides a binary EM coupling:

$$C_k = \begin{cases} +1 & \text{if level } k \text{ is active} \\ 0 & \text{if level } k \text{ is inactive} \end{cases}$$

The total coupling:  $C_{total} = \sum_{k=1}^{26} C_k / 26 \in [0, 1]$

At present universe:  $C_{total} \approx [SSq] = 0.57$  — exactly 15 of 26 levels active.

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## 5. Connection to Present-Day UQFF

The DPM initial conditions propagate to present physics as:

| DPM feature                      | Present UQFF parameter                            |
|----------------------------------|---------------------------------------------------|
| 26 spheres                       | 26-layer compressed gravity framework (SOURCE115) |
| SCm binding $E = 10^2 \text{ J}$ | $[SSq] = 0.57$ calibration                        |
| [UA] decay rate                  | $\_vac\_UA = 7.09 \times 10^{-3} \text{ J/m}^3$   |
| $+1/2$ / $-1/2$ states           | Ug fields vs. buoyancy fields                     |
| Resonance factor R               | $= 0.0005/\text{day}$ DPM coupling                |
| Binary encoding                  | $k\_ = 10^{113}$ suppression                      |

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## 6. Mathematical Derivation

### 6.1 DPM Gravitational Output

$$g_{DPM}(r, t) = R_{DPM} \times [SCm]^{26} \times e^{-\gamma t} = \frac{GMq_H H_s}{r^2} [SSq]^{26} e^{-\gamma t}$$

At  $t=0$  (Big Bang):  $g_{DPM} \approx GMq_H H_s / r^2 \approx g_{Newton}$  (DPM reduces to Newtonian at birth)

At  $t = 13.8 \text{ Gyr}$ :  $g_{DPM} \approx g_{Newton} \times (0.57)^{26} \times e^{-small} \approx 10^{-7} g_{Newton}$

This  $10^{-7}$  suppression is the origin of the UQFF quantum correction term  $a\_DPM = [SSq]$  g in the resonance MUGE equation.

## 6.2 DPM Momentum Term

$$F_{DPM,mom} = \frac{d}{dt} \left[ \sum_{k=1}^{26} m_k v_k \right] = \sum_{k=1}^{26} \frac{GMm_k}{r^2} C_k$$

This feeds directly into the  $F_{U_{Bi_i}}$  integral as  $F_{DPM,mom}$ .

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## 7. Observational Predictions

The DPM makes the following testable predictions:

1. **CMB power spectrum:** 26-sphere geometry  $\rightarrow$  specific  $l$ -multipole oscillations at  $l = 26, 52, 78$
  2. **Gravitational wave background:** 26D collapse  $\rightarrow$  specific GW frequency spectrum
  3. **Dark matter mass spectrum:**  $-1/2$  states  $\rightarrow$  SM-neutral particles at  $E = E_{SCm}/26 \approx 3.8 \times 10^{-10}$  J/particle
  4. **Vacuum energy fine-tuning:**  $\Lambda / F_{DPM} = 10^{-10}$  naturally from  $\exp(-t_{age})$  decay
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## 8. Conclusion

The DPM provides a pre-Big Bang origin model for the 26-dimensional structure underlying UQFF physics. The 26 spheres encode binary quantum levels, the  $[SCm]$  field stores  $10^{-2}$  J of binding energy, and the  $[UA]$  decay drives the transition from primordial high-energy to present low-energy vacuum. The model is mathematically self-consistent within the UQFF framework and makes testable predictions for the CMB, dark matter spectrum, and vacuum energy problem.

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**UQFF Parameters:**  $E_{SCm} = 10^{-2}$  J |  $[SSq] = 0.57$  |  $\Lambda_{UA} = 0.001 \text{ s}^{-1}$  | 26 spheres

**Class:** DPMModule | **Source:** `grok_share_b0a3dc1d.txt` L1871–2081

**Tags:** DPM, pre-Big-Bang, 26D, birth-model, vacuum-energy, CMB, inflation,  $[SCm]$ ,  $[UA]$